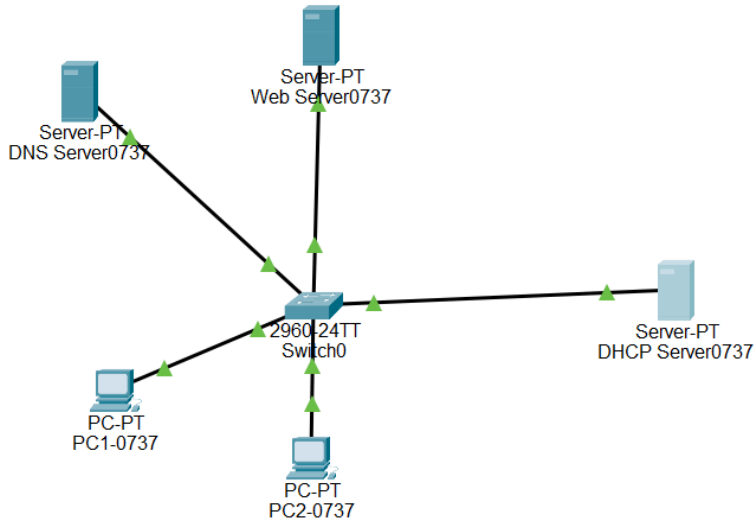


CN Lab 04 Submission

Name: Yash Raj

Roll Number: 24k-0737

Task 01:



DHCP Server:

DHCP Server0737

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: lab04

Default Gateway: 192.168.10.1

DNS Server: 192.168.10.2

Start IP Address : 172 168 10 1

Subnet Mask: 255 255 255 0

Maximum Number of Users : 255

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
lab04	192.168.1...	192.168.1...	172.168.1...	255.255.2...	255	0.0.0.0	0.0.0.0

DNS Server:

DNS Server0737

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type

Address

Add

Save

Remove

No.	Name	Type	Detail
0	googlefast	CNAME	googli
1	googli	A Record	172.168.10.5

Web Server:

Web Server0737

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

HTTP

HTTP ☒ On ☐ Off

HTTPS ☒ On ☐ Off

File Manager

	File Name	Edit	Delete
1	copyrights.html	(edit)	(delete)
2	cscoptlogo177x111.jpg		(delete)
3	helloworld.html	(edit)	(delete)
4	image.html	(edit)	(delete)
5	index.html	(edit)	(delete)

PC1:

PC1-0737

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 172.168.0.1

Subnet Mask 255.255.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

PC2:

PC2-0737

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 169.254.122.236

Subnet Mask 255.255.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

Packets Transmission:

DNS & PC:

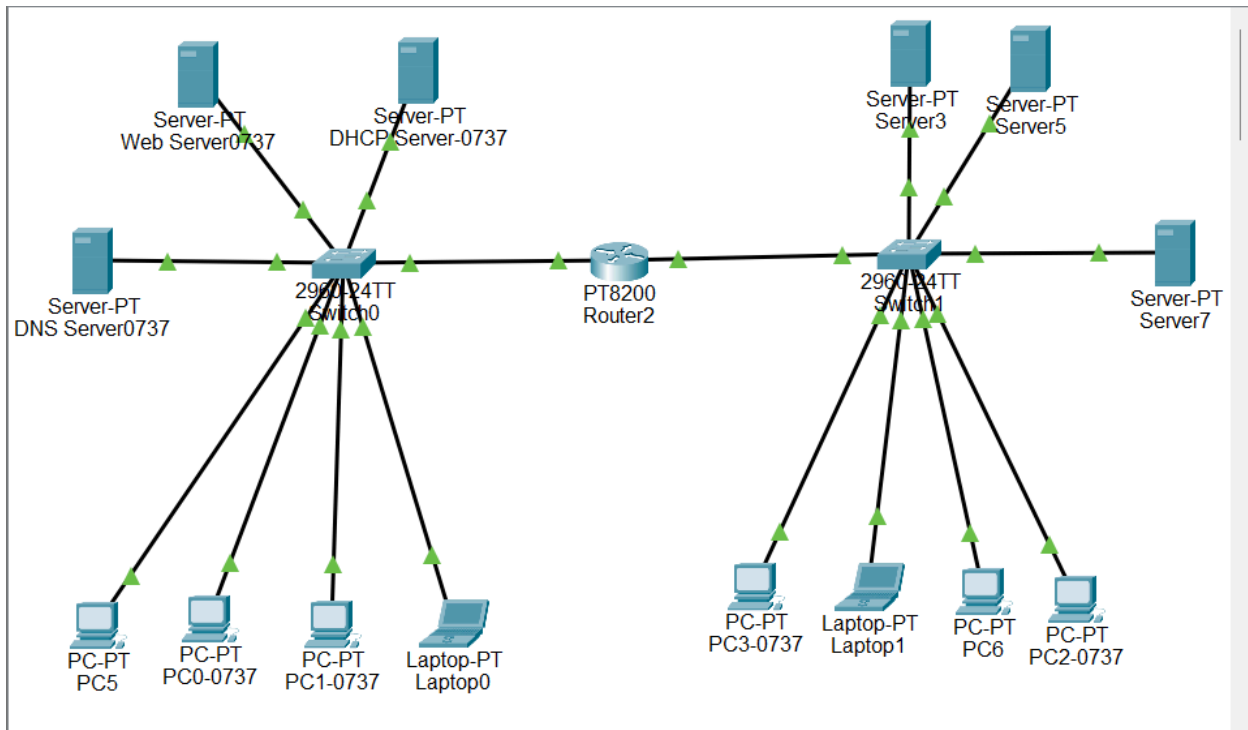
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1-0...	DNS Server...	ICMP		0.000	N	0	(edit)	(delete)

DNS & Web Server:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	DNS ...	Web Server...	ICMP		0.000	N	0	(edit)	(delete)

Task 02:

Topology:



Web Server:

Web Server-0737

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

HTTP

HTTP ☒ On ☐ Off

HTTPS ☒ On ☐ Off

File Manager

	File Name	Edit	Delete
1	cache-test.html	(edit)	(delete)
2	copyrights.html	(edit)	(delete)
3	cscoptlogo177x111.jpg		(delete)
4	helloworld.html	(edit)	(delete)
5	image.html	(edit)	(delete)
6	index.html	(edit)	(delete)

DNS Server:

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS**
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DNS

DNS Service ☒ On ☐ Off

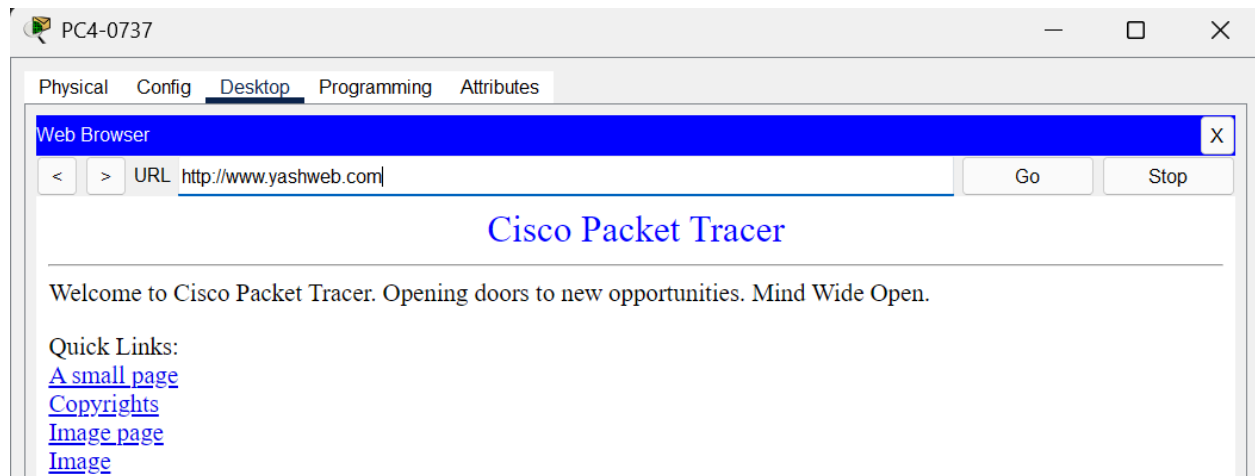
Resource Records

Name Type

Host Name

No.	Name	Type	Detail
0	dns.yashweb.com	A Record	192.168.10.2
1	yashraj	CNAME	www.yashweb.com
2	www.yashweb.com	A Record	192.168.10.3
3	yashweb.com	NS	dns.yashweb.com
4	yashweb.com	SOA	ServerName:dns.yash... MailBox :admin.yashweb.com Expiry :608400...

Testing:



Task 03: Which HTTP header sets a “best before” date for cached items?

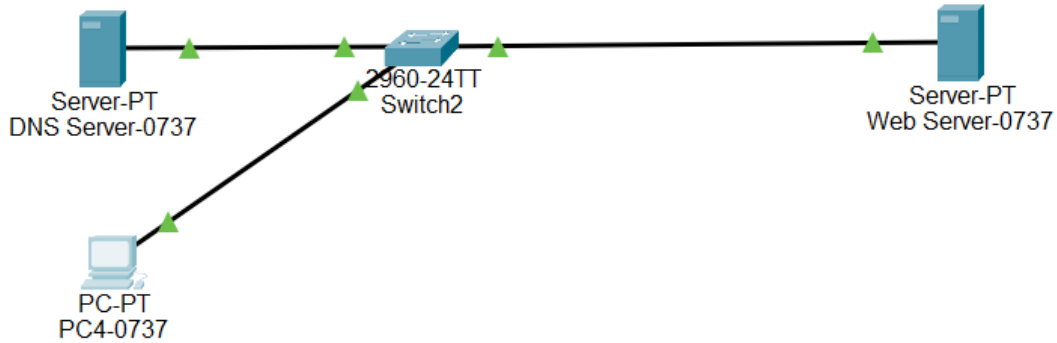
Answer: Expires header

Task 04: Are HTTP PUT and POST methods different?

Yes. POST is generally used to create a new resource where the server determines the URI, while PUT is used to create or completely replace a resource at a specific URI

Task 05:

Topology:



CNAME:

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS**
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type

Host Name

No.	Name	Type	Detail
0	dns.yashweb.com	A Record	192.168.10.2
1	yashraj	CNAME	www.yashweb.com
2	www.yashweb.com	A Record	192.168.10.3
3	yashweb.com	NS	dns.yashweb.com
4	yashweb.com	SOA	ServerName:dns.yash... MailBox :admin.yashweb.com Expiry :608400...

Testing:

Web Browser

< > URL

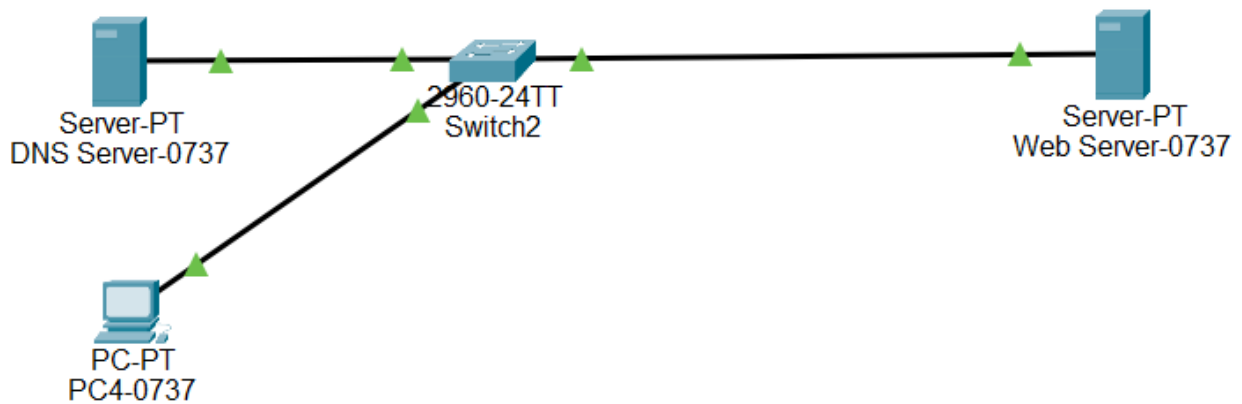
Cisco Packet Tracer

Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.

Quick Links:

- [A small page](#)
- [Copyrights](#)
- [Image page](#)
- [Image](#)

Task 06:



Web Server-0737

Physical Config Services Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

HTTP

☒ On ☐ Off

HTTPS

☒ On ☐ Off

File Manager

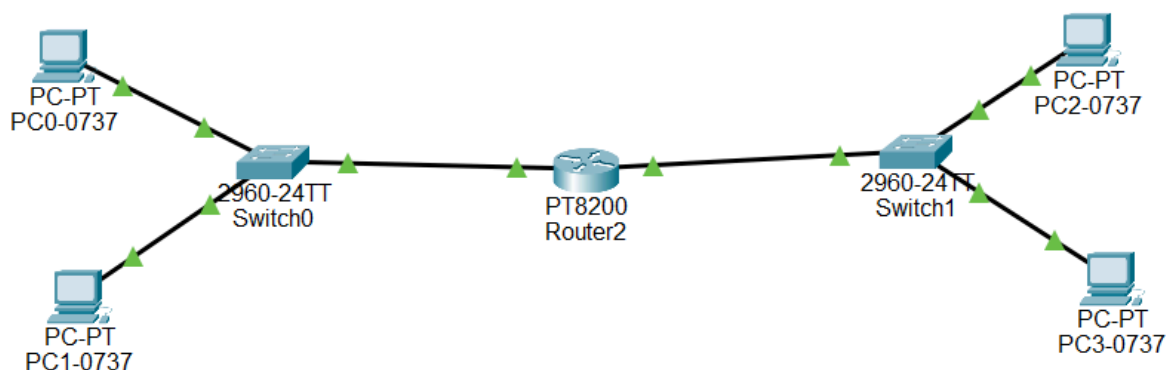
	File Name	Edit	Delete
1	cache-test.html	(edit)	(delete)
2	copyrights.html	(edit)	(delete)
3	cscoptlogo177x111.jpg		(delete)
4	helloworld.html	(edit)	(delete)
5	image.html	(edit)	(delete)
6	index.html	(edit)	(delete)

Web Browser

< > URL Go Stop

Hello

Task 07:



Testing:

```
C:\>ping 172.168.2.3
```

```
Pinging 172.168.2.3 with 32 bytes of data:
```

```
Request timed out.
```

```
Reply from 172.168.2.3: bytes=32 time=1ms TTL=127
```

```
Reply from 172.168.2.3: bytes=32 time<1ms TTL=127
```

```
Reply from 172.168.2.3: bytes=32 time<1ms TTL=127
```

```
Ping statistics for 172.168.2.3:
```

```
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>tracert 172.168.2.3
```

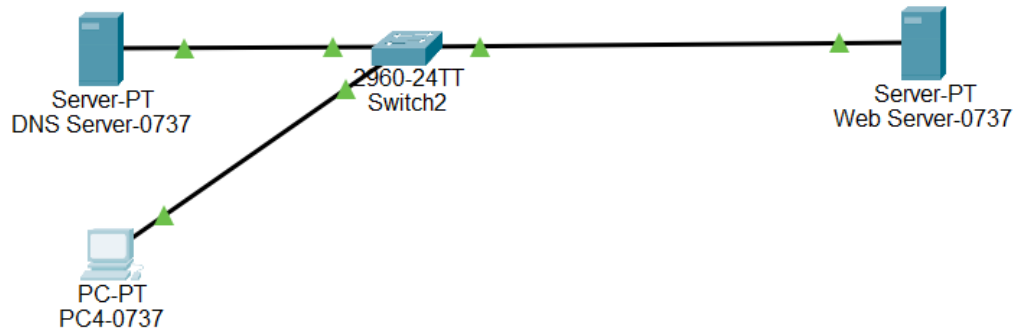
```
Tracing route to 172.168.2.3 over a maximum of 30 hops:
```

1	0 ms	1 ms	0 ms	172.168.1.1
2	1 ms	1 ms	0 ms	172.168.2.3

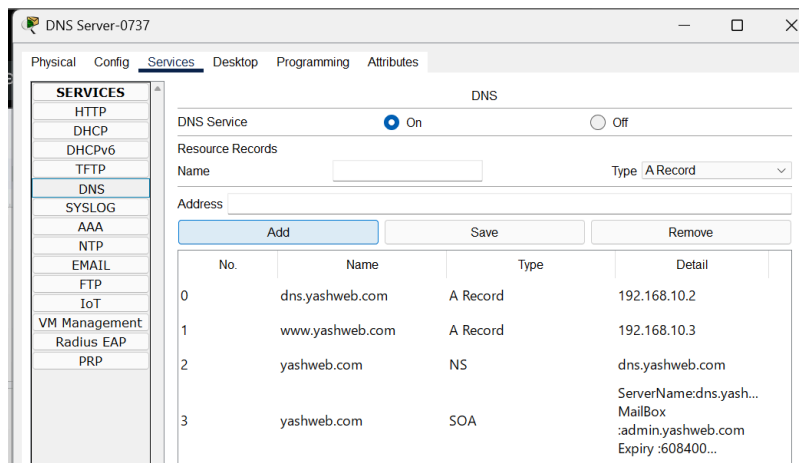
```
Trace complete.
```

Task 08:

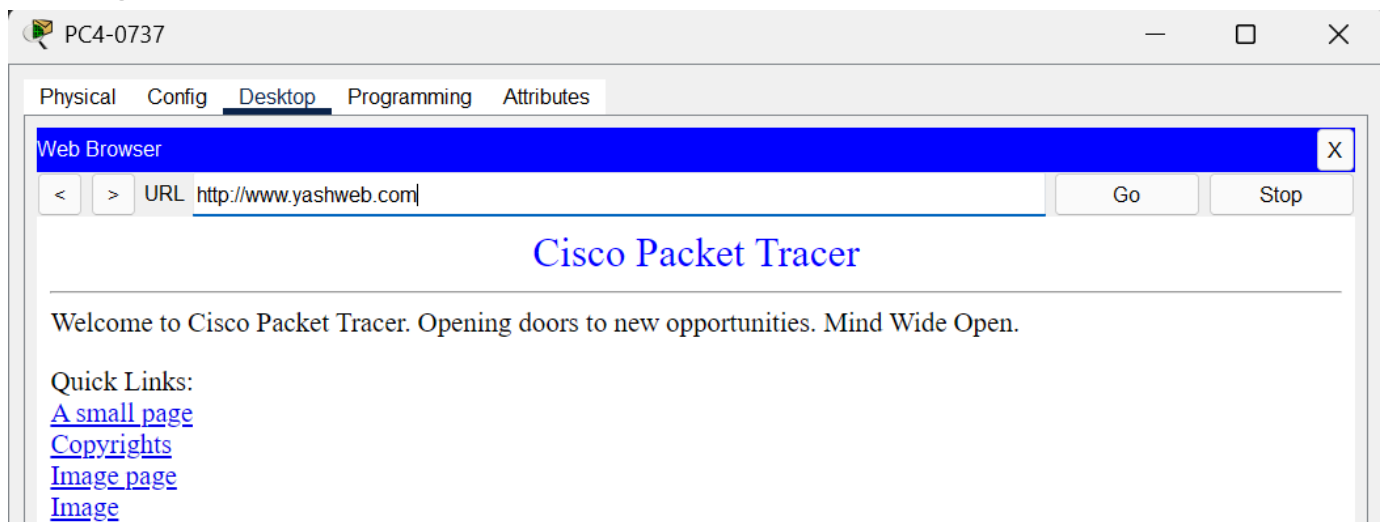
Topology:



DNS Server Configuration:



Testing:



CLI Testing:

```
Cisco Packet Tracer PC Command Line 1.0
```

```
C:\>ping 192.168.10.2
```

```
Pinging 192.168.10.2 with 32 bytes of data:
```

```
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.10.2: bytes=32 time=1ms TTL=128
```

```
Ping statistics for 192.168.10.2:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>ping 192.168.10.3
```

```
Pinging 192.168.10.3 with 32 bytes of data:
```

```
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
```

```
Ping statistics for 192.168.10.3:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>nslookup www.yashweb.com
```

```
Server: [255.255.255.255]
```

```
Address: 255.255.255.255
```

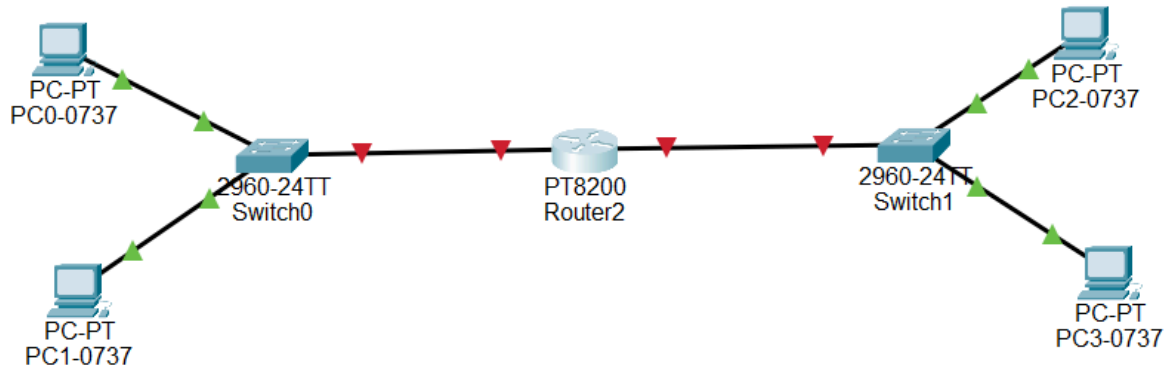
```
Non-authoritative answer:
```

```
Name: www.yashweb.com
```

```
Address: 192.168.10.3
```

Task 09:

Before:



Configuration:

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shut
Router(config-if)#exit
Router(config)#ip interface GigabitEthernet0/0/1
^
% Invalid input detected at '^' marker.
```

```
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shut
Router(config-if)#exit
Router(config)#end
Router#
```

```
Router#show ip interface brief
Interface          IP-Address      OK? Method Status  Protocol
GigabitEthernet0/0/0 192.168.1.1    YES manual up      up
GigabitEthernet0/0/1 192.168.2.1    YES manual up      up
GigabitEthernet0/0/2 unassigned      YES unset  down    down
GigabitEthernet0/0/3 unassigned      YES unset  down    down
Router#
Router#
```

After:

